

PRARABDHA PANDEY

Data Analyst | Marketing Analytics | SQL • Python • Power BI • Looker

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Published Patent Holder — NLP-based Automated Policy Query System (App No. 202521111197 A, Dec 2025) • Oracle
Certified: GenAI & Data Science Professional (Sep 2025)

PROFESSIONAL SUMMARY

Analytically driven B.Tech (AI & ML) graduate with hands-on experience in data analysis, ML model development, and building data pipelines. Skilled in SQL, Python, and data visualization platforms (Power BI, Looker) with a strong foundation in statistical analysis, customer segmentation, and campaign performance metrics. Proven ability to extract actionable insights from complex datasets, develop dashboards for stakeholders, and collaborate cross-functionally — directly aligned with HSBC's Data Analyst responsibilities in Branch & Retail Banking.

TECHNICAL SKILLS

Data Analysis & Reporting: SQL, Python (Pandas, NumPy, Matplotlib, Seaborn), Excel, Exploratory Data Analysis (EDA), Statistical Analysis

Visualization & BI Tools: Power BI, Looker, Google BigQuery, Dashboard Development, Performance Reporting

Machine Learning: Scikit-learn, XGBoost, Hugging Face Transformers, PyTorch, TensorFlow/Keras, NLP

Marketing Analytics: Customer Segmentation, Campaign Performance Metrics, Customer Engagement Analysis, CRM Data, ETL Pipelines

Databases & Infrastructure: MongoDB, Milvus (Vector DB), Semantic Search, Docker, Azure, Git/GitHub, FastAPI

PROFESSIONAL EXPERIENCE

Data & AI Analyst Intern | InspironLabs

Jan 2026 – Present

- Built end-to-end automated data pipelines for processing and analyzing large, complex document datasets using advanced chunking strategies, vector indexing, and semantic retrieval — reducing manual data processing effort significantly.
- Developed and deployed backend analytical services with structured query handling, request routing, and error-handling layers to integrate AI models with real-time data streams, closely mirroring CRM data integration workflows.
- Engineered multi-step analytical workflows with tool-calling and structured reasoning chains, enabling autonomous insight generation and reporting with minimal human intervention.
- Maintained detailed documentation of data pipelines, analytical methodologies, and reporting processes to ensure transparency and reproducibility across teams.

Machine Learning & Data Analyst Intern | Embrizon Technologies

Jul 2024 – Sep 2024

- Conducted Exploratory Data Analysis (EDA) on production datasets to surface feature patterns, data quality issues, and actionable insights critical to business decision-making.
- Developed and deployed predictive ML models (XGBoost, Scikit-learn) with hyperparameter tuning, achieving measurable improvement in classification accuracy (F1 score optimization) — directly applicable to customer segmentation and engagement analysis.
- Monitored and evaluated model performance metrics, identifying trends and opportunities for improving targeting strategies across datasets.
- Maintained production-ready experimentation workflows using Git, enabling collaborative versioning, reproducible experiments, and clean documentation for cross-functional teams.

KEY PROJECTS — DATA ANALYTICS & INSIGHTS

Customer Data Q&A & Analytics Backend — FastAPI, Milvus, Python, SQL, Docker

- Built a Dockerized microservice for document and customer data Q&A: integrated PDF/data parsing, vector storage (Milvus), and a FastAPI layer with full Swagger/OpenAPI documentation.

- Engineered semantic search capabilities enabling instant retrieval of patterns from large datasets — analogous to querying CRM platforms and digital marketing systems for campaign insights.
- Designed modular prompt templates and query-handling pipelines to generate structured, stakeholder-ready analytical outputs.

Multi-Source Data Intelligence & Reporting System — *LangGraph, LangChain, FastAPI, RAG, Python*

- Architected a multi-agent system for automated data triage: parallel evidence gathering (logs, metrics, structured data), hypothesis-verify loops, and auto-generated reports — translating raw data into actionable insights.
- Built RAG-powered pipelines over knowledge bases with metadata filtering; structured Pydantic outputs enabled consistent, report-ready data extraction comparable to campaign performance dashboards.
- Generated automated postmortem/analytical reports for each resolved data event, demonstrating ability to produce real-time reporting for business stakeholders.

Real-Time Data Streaming & Analytics Pipeline — *WebSocket, Python, FastAPI, Async Concurrency*

- Engineered a WebSocket-based real-time data pipeline for bidirectional streaming and inference, with async concurrent connection handling for low-latency data processing.
- Demonstrated ability to monitor real-time customer response rates and engagement metrics at scale — directly applicable to tracking campaign KPIs and customer engagement for HSBC's marketing initiatives.

PATENT

Computer-Implemented System for Automated Policy Query Response Using NLP — Issued Dec 19, 2025 | App No. 20252111197 A

- Designed an end-to-end NLP system covering policy data aggregation, text preprocessing pipelines, synthetic Q&A generation, and fine-tuned GPT-based architecture — demonstrating full-stack analytical product development from data ingestion to deployed inference.

EDUCATION

B.Tech — Artificial Intelligence and Machine Learning 2022 – 2026

Symbiosis Institute of Technology, Pune

10+2 — Science & Mathematics 2020 – 2022

Sainik School Ambikapur, Chhattisgarh

CERTIFICATIONS

- Oracle Certified Generative AI Professional (OCI) — Sep 2025
- Oracle Certified Data Science Professional (OCI) — Sep 2025